SU Jiajie | George

(+86)13531998622 joe_su.dg@foxmail.com

EDUCATION EXPERIENCE

The Chinese University of Hong Kong	Mechanical Automation Engineering	GPA: **3.4**/4.0	MSc(Project Grade: A-)	2024-2025
Dongguan University of Technology	Mechanical Design, Manufacturing and Automation	GPA: **3.8**/5.0	Bachelor(Rank:1/194)	2020-2024

Scholarship: Yang Zhenning Scholarship **National Scholarship, Ministry of Education, PRC**

PUBLICATION & PATENTS

PUBLICATION

[1] Jin X, Su J, Yan G, et al. A Low-Cost, 3D-Printable, Modular IMU-Based Rigid Data Glove [C]. 2026 IEEE International Conference on Robotics and Automation (ICRA) Workshop on Enabling Autonomy and Independence in Aging Societies through Advanced Robotics and AI, Vienna, Austria, June 5, 2026. (Accepted, Best Presentation Award).

[2] Su, J., Yu, Z., Hao, Z., So, R. H. Y., & Wei, Y. (2026). Anticipatory Auditory Cues for Mitigating Motion Sickness During Non-Driving-Related Tasks: The Roles of NDRT Modality and Cue Engagement. Manuscript in preparation (targeting Transportation Research Part F: Traffic Psychology and Behaviour).*

[3] Yu Z, Yang J, Tan J, et al. The simplification of the symptom checklist-90 scale utilizing machine learning techniques[J]. Journal of Affective Disorders, 2024, 366: 181-188.

[4] Yu, Z., Su, J., & Xue, J. Canonical correlation analysis of medical social support and quality of life in elderly depressed patients [J]. Journal of Guangdong Medical University, 1-11 [2025-10-30]. https://link.cnki.net/urlid/44.1731.r.20250418.1412.003. (in Chinese)

PATENTS

[1] Center for Artificial Intelligence and Robotics, Hong Kong Institute of Science & Innovation, Chinese Academy of Sciences; Institute of Automation, Chinese Academy of Sciences. Data glove for hand motion capture based on IMU sensors: CN202510644498.3 [P]. 2025-10-03. (in Chinese)

[2] Dongguan University of Technology. A picking device: CN202223593361.0 [P]. 2023-07-04. (in Chinese)

RESEARCH EXPERIENCE

Shenzhen University, School of Psychology ***Research Assistant*** 2026.03-present

Assisted and guided graduate students in equipment debugging, requirement-based coding, experimental data preprocessing and analysis, EEG stimulation experiments and data processing, as well as the development of VR environments and a six-axis motion platform.

Proj: 1. DTI-Based Brain Network Analysis for Motion Sickness Prediction: Applied machine learning to DTI-derived brain network data to predict motion sickness susceptibility and investigate differences in brain regions between adult and childhood motion sickness.

2. Effects of Predictability Cues and NDRT Intervention on Motion Sickness: Investigated the interaction between predictability cues and non-driving-related task (NDRT) intervention on motion sickness.

3. Supported multimodal motion-sickness prediction and patent/manuscript preparation by fusing 64-channel EEG, physiological, and 9-axis IMU data from 40+ subjects via modality-specific preprocessing and two-stage modeling.

Chinese Academy of Sciences Hong Kong Institute of Innovation ***Mechatronics Design*** 2024.09-2025.08

Completed data simulation for dexterous hand models based on teleoperated robotic arm control, assisted in improving hand-eye calibration and optimizing the robotic arm model structure. Automated the generation of large-scale 3D models using technologies such as web crawling, and ensured the accuracy of the robotic arm system.

Songshan Lake Materials Laboratory ***Research intern | Precision Medical Engineering Team*** 2023.04-2024.01

Conducted an extensive literature review, writes reports and reports on project progress.

Program software-defined radios to generate the required NMR signals, optimizing signal processing and data analysis efficiency, and Design and conduct experiments related to NMR to promote the progress of related research projects

PROJECT EXPERIENCE

AI-Based Trajectory Judgment and Generation for Micro-Robot Swarms 2025.01-2025.05

Zhang Li Team | Master's Project

Developed a real-time tracking and dynamic mapping framework integrating YOLOv8 and DeepSORT for continuous micro-robot swarm tracking in dense environments, and proposed a hot-update mapping algorithm that fuses historical trajectories with real-time segmentation masks to reduce trajectory fragmentation and generate smooth navigation heatmaps.

Mathematical Modeling Competition | Data Analysis and Mining Skills 2022.05-2023.02

American Collegiate Mathematical Contest in Modeling (MCM/ICM) | ***Honorable Mention*** 2023.02

The green GDP(GGDP) model is established through data mining and multi-factor analysis; responsible for data mining and establishing factor relationship analysis, in which code is written based on MATLAB.

National college students mathematical Contest in Modeling Guangdong area | ***Second prize*** 2022.09

Mainly studies the spatial structure scheduling model and writes the thesis; establish the positioning model based on geometric feature analysis.

May 1 Mathematical Modeling Contest | ***Second prize*** 2022.05

A purchasing strategy model is established, and the data are compared and analyzed by multiple planning; responsible for writing Lingo code to implement requirements planning; establish "autoregressive moving average time series model" to forecast demand.

A part monitoring system applied to the automated assembly line of superconducting cavity 2022.08-2023.02

Project host | National college students Innovation and Entrepreneurship Training Program

Completed the model structure design, proposed a dual-posture picking device and applied for a patent; utilized relevant control libraries in ROS to establish communication connections such as SSH, connected the server and client using the TCP/IP communication protocol, and integrated it into the software using WinForms, thereby improving the system's connectivity.

Fangke Energy -- China biomass thermal energy system customization expert 2022.05-2022.11

China College Students' "Internet+" Innovation and Entrepreneurship Competition Silver Award

Conduct on-site investigations to identify project advantages and improve plant operations; evaluate project execution to ensure timely completion and adherence to performance standards.

EXTRA-CURRICULAR ACTIVITIES

AutoGo Laboratory | Scientific Research Capability

Developed proficiency in C++, C#, Python, and Linux-based ROS, with hands-on experience in project-related engineering software. Participated in multiple research projects and contributed to one invention patent. Assisted graduate students in the National Superconducting Cavity Production Line Project by producing and optimizing 50+ engineering drawings.

College debate team | Communication and Analytical skills

Participated in multiple university debate competitions, developing strong critical thinking, logical reasoning, and argumentation skills, with experience in presenting and defending ideas effectively under pressure.

Asus Campus partner | Review and Team Collaboration Skills

Managed new media operations, including platform management and video editing, while participating in training programs on enterprise project organization, professional email etiquette, and forum operations. Planned and participated in an urban training camp, strengthening teamwork, communication, and personal presentation skills.

Other activities

- Joined the COVID-19 pandemic response volunteer team in Puning City and received the honorary title of "Outstanding Epidemic Prevention Volunteer."
- Conducted social practice research at revolutionary heritage sites, contributing to field research, documentation, and VR equipment-based photography and filming activities.
- Participated in a Smart Agriculture social practice project, conducting field investigations into local agricultural conditions in Yangshan, Qingyuan, and Hengli, Dongguan, and gaining firsthand insights into regional agricultural development.

SKILLS AND OTHERS

Languages : Cantonese (Basic), English(IELTS:6.5/6), Mandarin(Second Class B), Hakka(Native), Teochew(Native)

Software skills : Proficient in Matlab, SPSS, Lingo, MS Office, SolidWorks, CAD.

Other : Teacher qualification certificate(High school), Volunteer (time: More than 200h)